

Varnit Rawat

954-493-3350 | va501919@ucf.edu | linkedin.com/in/varnit-rawat | github.com/V4rnit | youtube.com/@Varnit-codes

EDUCATION

University of Central Florida

Orlando, FL

Bachelor of Science in Information Technology

Dec 2026

- Focus: Backend systems, Full-Stack Development, Enterprise Software Development and Mobile Development

PROJECTS

Secure Command-Line Shell | *Java, JWT, Multithreading, File I/O, CLI, Linux*

Oct 2025

- Built a modular **Java**-based command-line shell with support for custom commands, pipelines, and extensible execution flow.
- Introduced **JWT**-based session management early in the execution pipeline, cutting unauthorized command attempts by **55%**.
- Improved interactive performance by parallelizing command execution with **multithreading**, resulting in noticeably lower latency **35%** during heavy usage.

AlgoPath | *HTML, CSS, JavaScript, React, Tailwind CSS, Node.js, Firebase*

July 2024

- Built a full-stack learning platform using **React** and **Node.js** with user authentication, persistent progress tracking, scalable data models, and successfully serving **100+** students.
- Designed mobile-first, responsive UI components with **React** and **Tailwind CSS**, improving mobile user retention by **25%**.
- Implemented backend APIs and **Firebase** listeners to synchronize user progress in near real-time, increasing daily active interactions by **20%**.

CodeClarity | *React, Node.js, Express, MongoDB, REST APIs, JavaScript, HTML/CSS*

May 2025

- Simulated real interview scenarios covering **20+** common data structures and algorithm patterns, enabling self-assessment and structured reasoning feedback.
- Built a full-stack web application generating **100+** unique code snippets for users to practice algorithm explanation and analyze time and space complexity.
- Developed backend with **Node.js** and **Express** to handle concurrent submissions, persisting results in **MongoDB**, achieving an interactive experience with **<200ms** average response time.

EXPERIENCE

Teaching Assistant — Data Structures & Algorithms

Aug 2025 – Present

University of Central Florida

Orlando, FL

- Assisted **140+** students during lab sessions by debugging code and explaining core DSA topics such as arrays, trees, and sorting algorithms.
- Created eight unique corner-case test cases, improving code quality and saving students approximately **20+** of debugging time per week.
- Improved grading efficiency by **50%** by developing automated Bash scripts to grade programming assignments, ensuring accuracy and consistency.

Workshop Instructor

Aug 2025 – Present

Knight Hacks

Orlando, FL

- Managed workshops for **40+** students on coding strategies for technical interviews.
- Guide students on effective communication and coding approaches during interviews, resulting in **20%** higher mock interview success rates.
- Collaborated with peers in weekly planning sessions, iterating on curriculum based on student feedback and achieving a **30%** improvement in outcomes.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, Bash

Frameworks: React, Node.js, JUnit, WordPress, Spring Boot

Developer Tools: Git, Docker, Google Cloud Platform, VS Code, PyCharm, IntelliJ, TMUX, Vim, NeoVim

Concepts: Multithreading, CLI, REST APIs, JWT Authentication

Libraries & Tools: JWT (jsonwebtoken)